```
  else if(SupplyState == LOW && YellowState==HIGH)
  //WATER LEVEL BELOW 75% AND SUPPLY IS OFF
{
    digitalWrite(9, LOW);
    digitalWrite(8, HIGH); // TURN ON PUMP

    lcd.clear();
    lcd.print("PUMP: ON");
    lcd.setCursor(0,1);
    lcd.print("SUPPLY OFF");
  }
  else
  {

    digitalWrite(9, LOW);
    digitalWrite(8, LOW);

    lcd.clear();
    lcd.print("PUMP: OFF");
    lcd.setCursor(0,1);
    lcd.print("SUPPLY OFF");
  }
delay(2000);
}
```

Fig. 5: Arduino code part 3

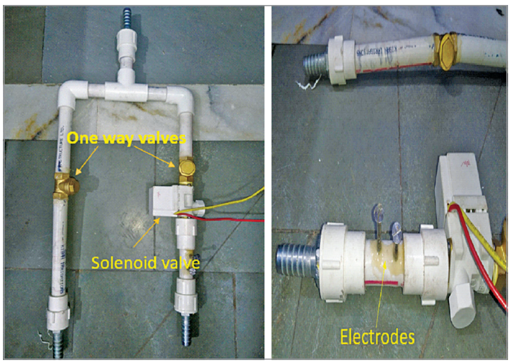

Fig. 6: Mechanical pipe and valve assembly

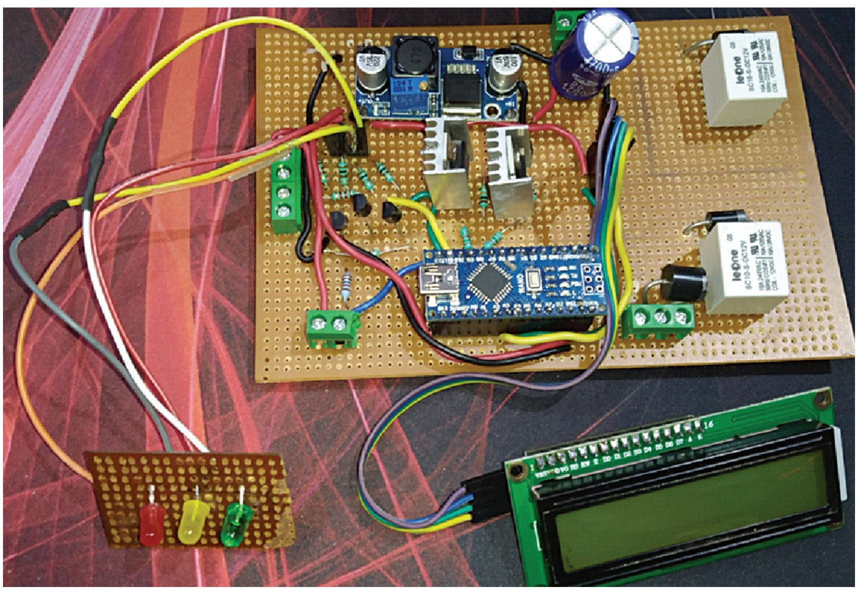

Fig. 7: Author's prototype of the circuit

with I2C/IIC interface (MOD2), three BC548 transistors, two IRF530 MOSFETs, and a relay.

The availability of municipal supply is detected by the two electrodes drilled into the pipe and connected to Arduino. The gap between the electrodes acts as a hindrance to the flow of current (as resistance). When the municipal water is available, the current starts flowing through the electrodes, which is detected by the microcontroller in Arduino.

The flow of water from the municipal supply is controlled through a solenoid valve and the water pump is used to fill the water from underground storage tank to rooftop tank. Two one-way valves are used to prevent reverse flow of water. The two water supplies are merged to a single outlet through a 'T' connector.

The water level is determined by the three BC548 BJT transistors that act as switches. The bases of the BJTs are connected to the electrodes dipped into the tank at different levels (50%, 75%, and 100%) through resistors. When the water level reaches an electrode then the corresponding transistor turns on and thus the LED connected to the transistor also turns on. The collector pins of two transistors are also connected to the digital pins of the Arduino to further define the logic of the circuit.

The 16x2 LCD connected to the Arduino Nano through I2C displays the status of water supply and water pump.

The solenoid valve is controlled via a relay. The relay is controlled by the IRF530 MOSFET. The gate pin of the MOSFET is connected to the digital pin D9 of Arduino Nano through a resistor. The resistor connected between gate of the MOSFET and the ground acts as a pull-down resistor. A flyback diode is connected in parallel to the coils of the relay to protect the MOSFET from high voltage created by collapsing magnetic fields.

The water pump can be directly controlled through the relay but due to high inrush current of the pump, the contacts of the relay can easily get damaged. Hence, a contactor similar to a relay is used, but it operates on 220V AC and has a bigger and heavier metal contacts that do not damage easily. The contactor is controlled through the same relay-MOSFET system as explained above. The gate pin of the MOSFET is connected to digital pin D8 of Arduino Nano.

The Arduino Nano is powered with a 5V input, which is achieved through a buck converter connected to the 12V supply. A 4700µF capacitor is connected to the 12V power supply to act as a low-pass filter.

Software

Arduino Uno IDE is used here. In the Arduino code, the behaviour of the hardware components is defined based on the methodology mentioned earlier.